# ApexPlanet Task 4 – Hypothesis Testing Summary

## 1. Project Overview

This project is part of my Data Analytics Internship at ApexPlanet Software Pvt. Ltd.

The objective is to analyze sales data, identify business insights, and perform statistical hypothesis testing using Python.

## 2. Dataset Information

- Total records: 1,000
- Total sales: INR 13,93,99,439.65
- Missing Age values: 20
- Missing City values: 13
- Duplicate records: 0

## 3. Business Insights

- Electronics recorded the highest sales.
- Patna was the highest-performing city.
- Total sales across all records were INR 13,93,99,439.65.

## 4. Hypothesis Testing

**Null Hypothesis (H0):** There is no significant difference between the average sales of male and female customers.

**Alternative Hypothesis (H1):** There is a significant difference between the average sales of male and female customers.

Statistical method: Welch's Independent Samples T-Test.

- Female average sales: INR 136,883.21
- Male average sales: INR 141,807.34
- T-statistic: 0.6826
- P-value: 0.4950

- Significance level: 0.05

**5. Result and Conclusion**

Since the p-value (0.4950) is greater than 0.05, we fail to reject the null hypothesis.

There is insufficient statistical evidence to conclude that average sales differ between male and female customers.

**6. Tools Used**

- Python
- Pandas
- NumPy
- SciPy
- Matplotlib

**7. Conclusion**

This project helped me develop practical skills in data analysis, business storytelling, statistical hypothesis testing, and data visualization.